Database Forensics

How Data Solves Crimes in Large Organizations

A 5 - slide guide to investigating data at scale with SQL and audit logs

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Conversion Systems: - Unit II

Unsigned integer representation: is a binary number representation system in which all bits are used to represent the magnitude of a non-negative integer
with no sign bit
and a value range from 0 to $(2^n - 1)$.

Unsigned binary number

- Uses only 0 and 1
- Represents only positive values (including zero)
- Has no sign bit
- All bits are used for the value

- Think of it like: “no negatives allowed”

Uses

- **Unsigned tells us how to INTERPRET the same 0s and 1s as a number.**
- **Unsigned = meaning of that data.**
- **To represent only positive numbers (age, count memory address, pixel values (0–255))**
- **To get a larger positive range**
- **To avoid confusion in calculations**

- Unsigned binary number = the actual bit pattern
- Unsigned integer representation = the rule used to interpret that bit pattern

What Is Unsigned Integer Representation?

Unsigned binary number e.g; 11111111
Unsigned Integer representation e.g; 255

- We use binary (0s and 1s)
- We assume the number is ALWAYS positive or zero
- No sign bit is used
- All bits are used to store the value

NOTE:

- It is NOT a special type of bit
- It is NOT extra hardware
- It is NOT a new system of writing numbers
- It is ONLY a rule of interpretation

Unsigned integer representation = binary numbers interpreted as positive whole numbers only

PLACE VALUE SYSTEM

1. NIBBLE (4 BITS)

0 to $(2^4 – 1)$

THE RANGE OF VALUES = 0 to $(2^n – 1)$.

n is any positive whole number from 0

for n = 4

n= 0, 1, 2, 3, 4

1 2 4 8 16

range for a nibble

16 - 1 = 15

range will be 0-15

two states of the computer

ON = 1

OFF = 0

USE THE 8 4 2 1 rule

0000 = 0

1111 = 8 + 4 + 2 + 1 = 15

2. BYTE (8 BITS)

0 to $(2^8 – 1)$

(n=8)

n = 0 , 1, 2, 3, 4, 5,6,7,8

1, 2, 4, 8, 16, 32, 64, 128, 256

256-1 = 255

range for a byte is 0-255

state all OFF 128 64 32 16 8 4 2 1 rule

00000000 = 0

all ON

11111111 = 1 + 2 + 4 + 8 + 16 + 32 +64 +128 = 255

255 is the highest whole number when all the bits are on in a byte

8 bits = It actually means 8 binary positions forming one number

UNSIGNED VALUE RANGE

For n bits, the range is:

0 to $(2^n - 1)$

Examples:
3 bits → 0 to 7
4 bits → 0 to 15
8 bits → 0 to 255

Use 8 4 2 1
0101_2 = 0 + 4 + 0 + 1 = 5
0000_2 = 0

1111_2 = 8 + 4 + 2 + 1 = 15

00000001 = 1
10000000 = 128
00001111_2 = 15
10101010_2 = 128 + 0 + 32 + 0 + 8 + 0 + 2 + 0 = 170

11111110_2 = 255 - 1 = 254

128 64 32 16 8 4 2 1

In unsigned binary, every bit increases value — there is NO negative side.

KEY TAKEAWAY

- Each bit represents a power of 2
- Turning a bit ON adds its value
- Unsigned binary cannot represent negative numbers

- Summary

- Only positive numbers
- No sign bit
- Range depends on number of bits
- Used in basic computer data representation

- "Unsigned bit" is NOT a thing
- There is no such thing as an "unsigned bit."
- What you actually have is:

- Unsigned binary number (a way of interpreting bits)

- Not a special bit.

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